

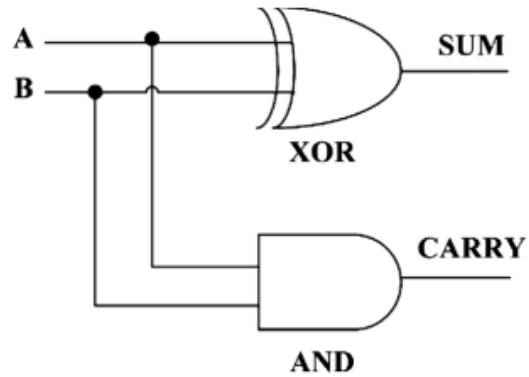
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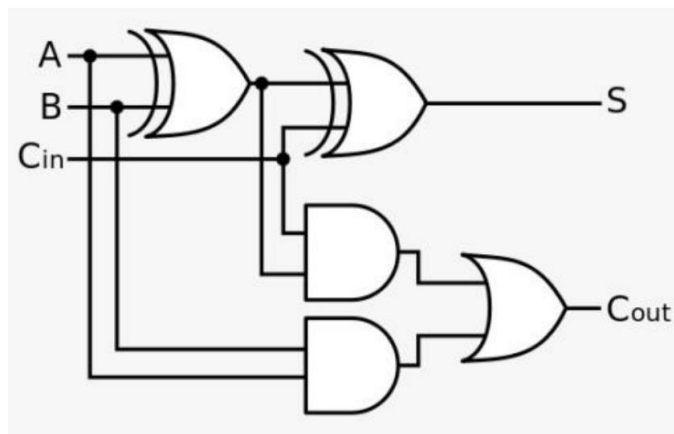
Practice #2: Boolean equation with Verilog in Cadence

Describe two basic combinational modules: one Half-Adder and one Full-Adder. For each module simulate to verify the expected behavioral in all cases.

HDL code



```
1 //Verilog HDL for "Verilog", "Half_Adder" "functional"
2
3 module Half_Adder (A,B,Sum,Carry);
4
5     input  A,B;
6     output Sum,Carry;
7
8     assign Sum = A ^ B;
9     assign Carry = A & B;
10
11 endmodule
```



```
1 //Verilog HDL for "Verilog", "Full_Adder_TB" "functional"
2
3 module Full_Adder(A,B,Cin,S,Cout);
4
5     input  A,B,Cin;
6     output S,Cout;
7
8     assign S = ((A ^ B) ^ Cin);
9     assign Cout = (((A ^ B) & Cin) | (A & B));
10
11 endmodule
```

Test bench codes

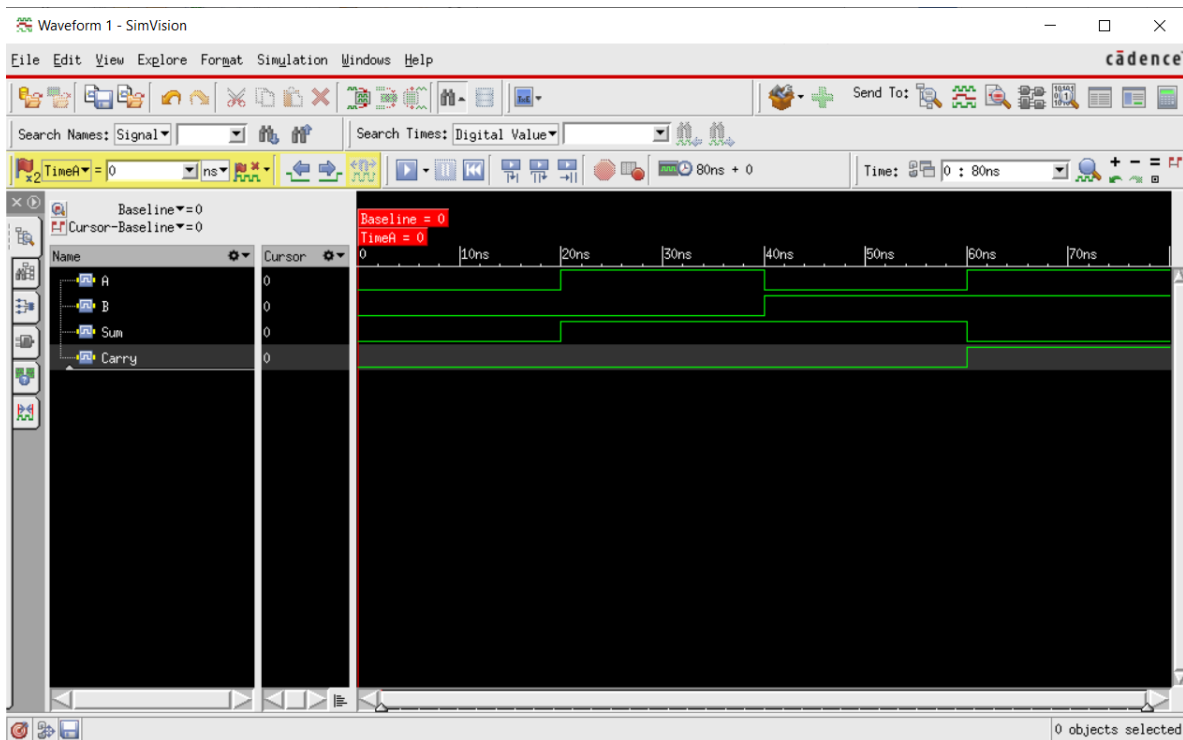
```
1 //Verilog HDL for "Verilog", "Half_Adder_tb" "functional"
2
3 module Half_Adder_tb();
4
5     reg A,B; //Creamos registros a nuestras entradas
6     wire Sum; //Creamos cables para nuestras salidas
7     wire Carry;
8
9     Half_Adder uut (A,B,Sum,Carry); //Creamos el modulo de la intancia del archivo que le haremos su TB
10
11     initial begin
12
13         A = 0;
14         B = 0;
15         #20; //Espera 20nS entre cambios
16
17         A = 1;
18         B = 0;
19         #20; //Espera 20nS entre cambios
20
21         A = 0;
22         B = 1;
23         #20; //Espera 20nS entre cambios
24
25         A = 1;
26         B = 1;
27         #20; //Espera 20nS entre cambios
28
29         $display("Test completed");
30
31     end
32
33 endmodule
```

```
1 //Verilog HDL for "Verilog", "Full_Adder_TB" "functional"
2
3 module Full_Adder_tb();
4
5 integer i;    //for cycle variable
6 reg  A,B,Cin;
7 wire S;
8 wire Cout;
9
10 Full_Adder uut (A,B,Cin,S,Cout);
11
12 initial begin
13     $dumpfile("Full_Adder_tb.vcd");
14     $dumpvars(0,Full_Adder_tb);
15
16     $display("\tA\t\tB\t\tC\t\tS\t\tCout");
17
18     for (i=0; i <= 7; i=i+1 ) begin
19
20         A = i[2];
21         B = i[1];
22         Cin = i[0];
23         #10;
24
25         $display("\t%d\t\t%d\t\t%d\t\t%d\t\t%d",A,B,Cin,S,Cout);
26
27     end
28
29     $display("Test completed");
30
31 end
32 endmodule
```

Simulation

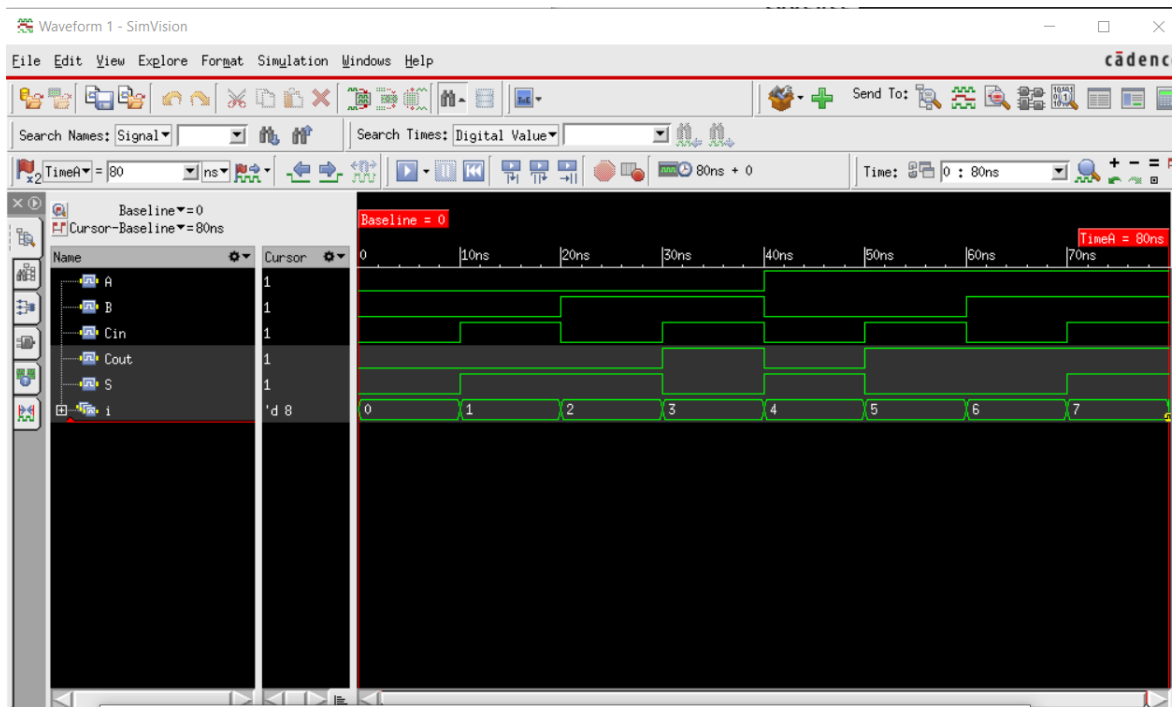
Half Adder truth table			
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

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Full Adder truth table				
A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

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Composer NC-Verilog Integration Log File

File Edit View Help

xcelsius> run

A	B	C	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Test completed

xcelsius>

xcelsius> run

CODES

//Verilog HDL for "Verilog", "Half_Adder" "functional"

```
module Half_Adder (A,B,Sum,Carry);
```

```
    input A,B;
    output Sum,Carry;
```

```
    assign Sum = A ^ B;
    assign Carry = A & B;
```

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endmodule

//

//Verilog HDL for "Verilog", "Half_Adder_TB" "functional"

module Half_Adder_tb();

reg A,B;

wire Sum;

wire Carry;

Half_Adder uut (A,B,Sum,Carry);

initial begin

A = 0;

B = 0;

#20;

A = 1;

B = 0;

#20;

A = 0;

B = 1;

#20;

A = 1;

B = 1;

#20;

\$display("Test completed");

end

endmodule

//

//Verilog HDL for "Verilog", "Full_Adder_TB" "functional"

module Full_Adder(A,B,Cin,S,Cout);

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```
input A,B,Cin;
output S,Cout;

assign S = ((A ^ B) ^ Cin);
assign Cout = (((A ^ B) & Cin) | (A & B));

endmodule

/////////////////////////////////////////////////////////////////
//Verilog HDL for "Verilog", "Full_Adder_TB" "functional"

module Full_Adder_tb();

integer i; //for cycle register
reg A,B,Cin;
wire S;
wire Cout;

Full_Adder uut (A,B,Cin,S,Cout);

initial begin

    $display("\tA\t\t\tB\t\t\tC\t\t\tS\t\t\tCout");

    for (i=0; i <= 7; i=i+1 ) begin

        A = i[2];
        B = i[1];
        Cin = i[0];
        #10;

        $display("\t%d\t\t\t%d\t\t\t%d\t\t\t%d\t\t\t%d",A,B,Cin,S,Cout);

    end

    $display("Test completed");

end

endmodule
```